Machine Learning

Naïve Bayes- Week 2

Objectives

- Exam and Project Deadlines
- Bayes Rule
- Formulation of Naïve Bayes

Where to find me?

Office:

Science Hall 250

Office Hours:

Weds and Thurs 11:00 AM - 1:00 PM EST or by appointment

Exams (60%)

Test	Date	Time	Details
Test 1 (25%)	Oct 3rd, 2025	During lecture hours	materials covered till Sept. 30th
Test 2 (25%)	Nov 14th , 2025	During lecture hours	materials covered till Nov 11th
Final (50%)	Dec 12th, 2025	10:15 AM –12:15 PM	Everything covered in the course

Project (30%)

Task: Classification/Regression (or other tasks) on a novel dataset.

You need to train/test at least **3 algorithms** taught in the class on this dataset. This is the minimum requirement for undergrads. For graduate students, you need to come up with an **additional algorithm** that we have not discussed in class for your project and explain it in detail in your project report with proper citations.

Final deliverable is on December 5th by 5:00 PM EST). You need to submit a project report in the format attached in BrightSpace along with codebase, either as Github link (preferred) or OneDrive.

Datasets: Check Kaggle

Project – Intermediate Deliverables

<u>Sep 26th, 2025</u>: what is your project about? What datasets are you using? Motivation for doing this project? Some preliminary literature review. **Deliverable**: Email me a document with all these details. After you submit these steps, I'll schedule meetings to go over the idea.

<u>Oct 24th, 2025</u>: By this time, since your project is finalized, you should have completed the Motivation, Literature Review/Related Work, Methods of your project report. In terms coding, the following steps should be complete

- Data cleaning and extraction
- Training and testing at least 2 ML method (3 for grad students)
- Some preliminary results on the training and testing.

Deliverable: Project report with Motivation/ Introduction, Related Work, Methods and Preliminary results; code base either as a GitHub link or one drive folder shared with me.

Project – Intermediate Deliverables

Nov 7th, 2025: All codes should be completed by this time. You may only be working on the project report and presentations. You will need to discuss the dataset, some related work, ML methods you use, results.
Deliverables: Codebase, project presentation.
Deadline for presentation submission: Nov 7th

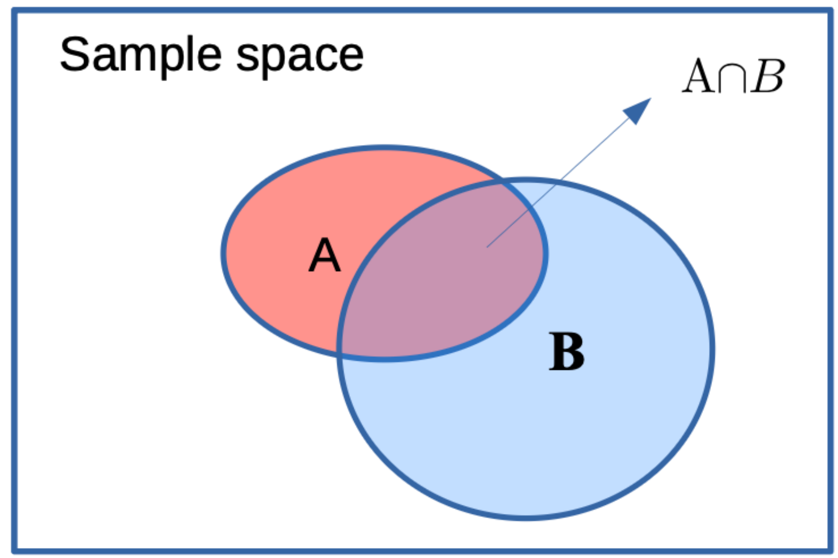
Presentation Approvals will be sent by Nov 14th. **Presentations starts** on **Nov 18th - Dec 5th**, right after with class lectures.

Final deliverable (December 5th by 5:00 PM EST): Project report in the format attached in BrightSpace along with codebase, either as Github link (preferred) or OneDrive.

Reviewing Concepts

- The union of the two events is A or B, $A \cup B$
- The intersection of the two events, $A \cap B$
- Union is an **OR** operation; intersection is an **AND** operation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



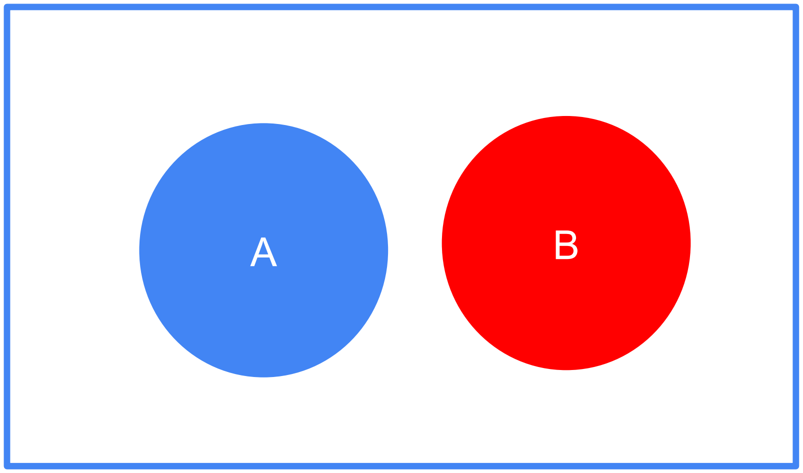
***The rectangle is a whole sample space. A and B are just two events*

Mutually Exclusive Events

- The events cannot occur together. i.e., there is no intersection between the events. E.g. Flipping a coin.

- For mutually exclusive events,

$P(A \cup B) = P(A) + P(B)$ since $P(A \cap B) = 0$



***The rectangle is a whole sample space. A and B are just two events*

Independent Events

- The occurrence of one event doesn't affect the occurrence of the other event.
- We cannot visualize independent events through Venn diagram!
- For independent events, $P(A \cap B) = P(A) * P(B)$

Conditional Events

- In situations where we have some partial information of an outcome of an experiment, but not the whole information, we have what is called a **conditional** event.

Suppose we know that the dice toss resulted in an even number (event A). We may be interested in the probability of of the toss being greater than 2 (event B). We use the notation:

$P(B|A)$ → "Probability of B given A" = $P(A \cap B)$ / $P(A)$

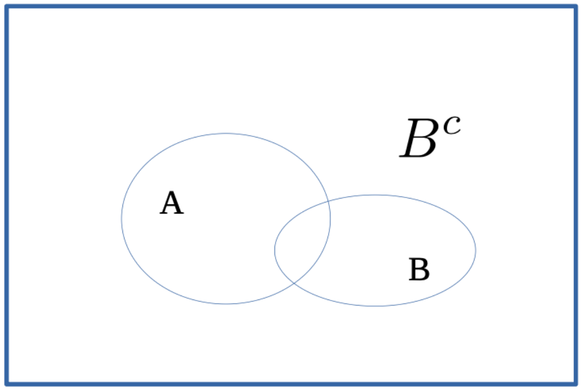
Since A already, the effective set of outcomes, or sample space, has now shrunk to {2, 4, 6}. Of these, only 4 and 6 are greater than 2, resulting in P(B|A) = 2/3

Conditional Events

- If A and B are independent, then $P(B|A) = P(B)$ and $P(A|B) = P(A)$
- If A and B are mutually exclusive, $P(B|A) = P(A|B) = 0$

The concept of total probability

It is often useful to express the probability of some event, A, in terms of the probability of some other event, B. By looking at the Venn diagram, we can see how to express $P(A)$ in terms of conditionals:



$$P(A) = P(A \cap B) + P(A \cap B^c)$$

Since $P(A \cap B) = P(A|B) * P(B)$ and $P(A \cap B^c) = P(A|B^c) * P(B^c)$, we can write:

$$P(A) = P(A|B) * P(B) + P(A|B^c) * P(B^c)$$

This expression is called **law of total probability**.

Bayes Rule

Bayes theorem is a fundamental statistical tool used in many scientific disciplines. It is intuitively simple to derive, and states that one conditional probability can be described in terms of the inverse conditional probability scaled by a factor.

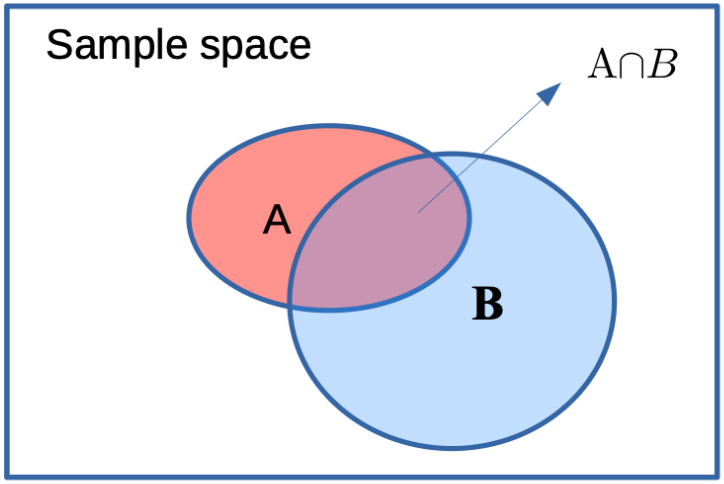
Consider two events A and B in some sample space. We have seen that the conditional probability $P(A|B)$ can be written as

$P(A|B) = \frac{P(A \cap B)}{P(B)}$ or, $P(A \cap B) = P(A|B) * P(B)$

Similarly, $P(B \cap A) = P(B|A) * P(A)$ and since $P(A \cap B) = P(A \cap B)$, we can write:

$P(A|B) * P(B) = P(B|A) * P(A)$ or

$P(A|B)= \frac{P(B|A) * P(A)}{P(B)}$ This is the mathematical statement of Bayes' Theorem



Bayes Rule

In general, given a class variable, y and some features, x_1 through x_n, using Bayes' Theorem we can write the following:

$$\underbrace{P(y|x_1, x_2, \dots, x_n)}_{\text{Posterior}} = \frac{\overbrace{P(x_1, x_2, \dots x_n|y)}^{\text{Likelihood}} * \overbrace{P(y)}^{\text{Prior}}}{\underbrace{P(x_1, x_2, \dots x_n)}_{\text{Evidence}}}$$

In practice, there is interest only in the numerator of that fraction, because the denominator does not depend on y and the values of the features are given, so that the denominator is effectively constant.

Naïve Bayes

$$P(y|x_1, x_2, \ldots, x_n) = \frac{P(x_1,x_2,\ldots,x_n|y) * P(y)}{P(x_1,x_2,\ldots,x_n)} \longrightarrow (1)$$

Using the assumption that every feature is independent of every other features , given a specific outcome or category, we can re-formulate (1) as:

$$P(y|x_1, x_2, \ldots, x_n) = \frac{P(y)\prod_{i=1}^{n} P(x_i|y)}{P(x_1,x_2,\ldots,x_n)}$$

Therefore, $\hat{y} = argmax\ P(y)\prod_{i=1}^{n} P(x_i|y)$

We use Maximum A Posteriori (MAP) estimation to estimate $P(y)$ and $P(x_i|y)$; $P(y)$ is equivalent to the relative frequency of class, y in the training set.

ANY QUESTIONS?